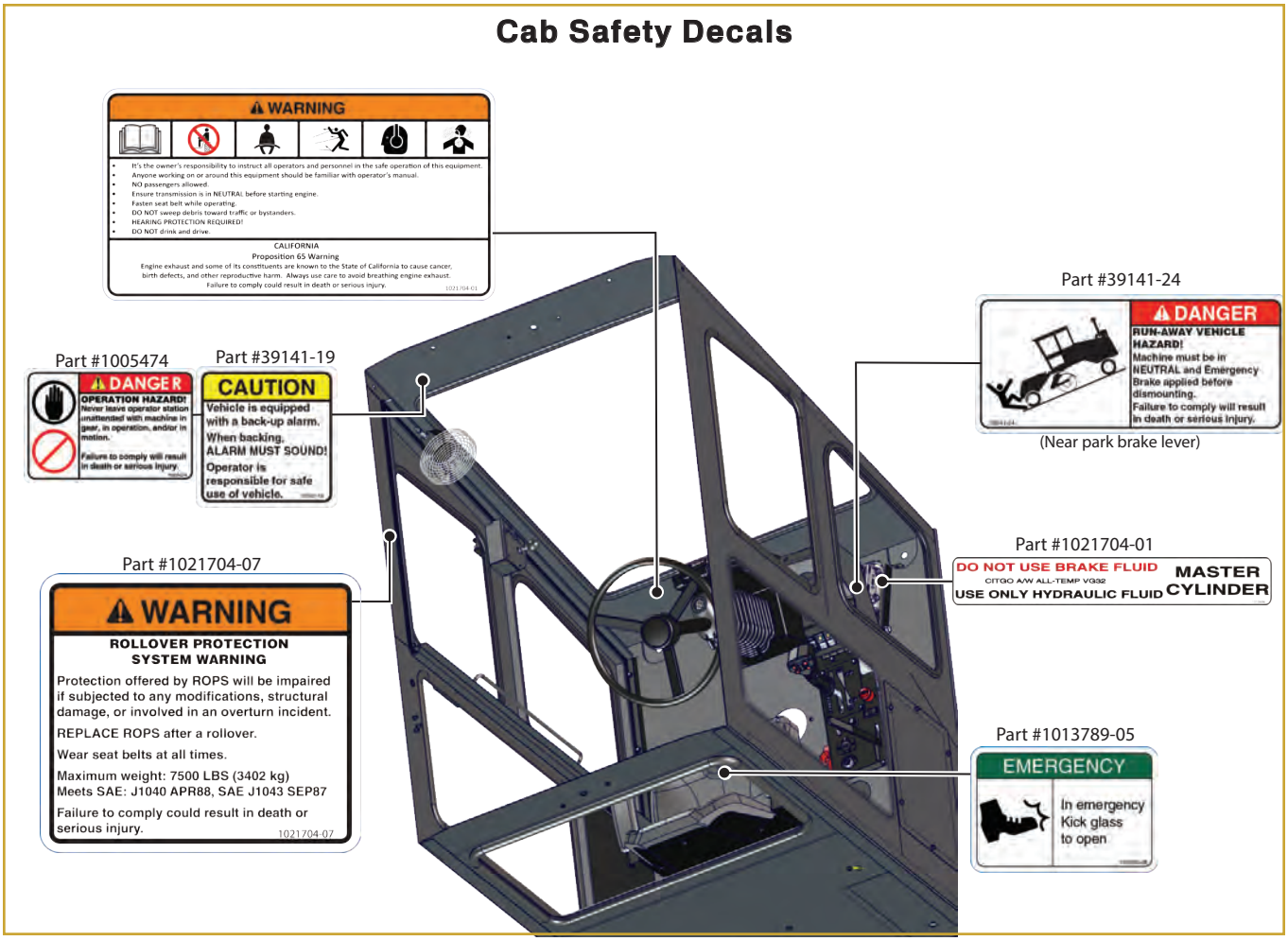
Cab Safety Decals



Safety Decals Care

- Keep safety decals and signs clean and legible at all times.
- Replace decals and signs that are missing or become impossible to read.
- Become familiar with the content and position of each safety decal.
- When replacing parts that display a safety decal, ensure that the new part is fitted with the correct decal.
- Obtain replacement safety decals or signs from your authorized dealer.

Decal Application

1. Be sure the installation area is clean and dry. Use hot, soapy water to clean the surface.
2. Thoroughly dry the surface.
3. Measure and test fit before removing decal paper backing.
4. Remove the smallest split backing paper and align decal over the specified area, carefully pressing exposed portion into place.
5. Slowly remove remaining backing and carefully smooth into place.
6. Small air pockets can be pierced with a pin and smoothed using a piece of decal backing.

Section 2

INFORMATION AND SPECIFICATIONS

2

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TWO YEAR LIMITED WARRANTY

Limited Warranty

If a manufacturing defect in factory supplied materials or factory workmanship is found and the authorized LeeBoy dealer ("Dealer") is notified during the warranty period, LeeBoy will be responsible for repairing or replacing any part or component of the unit or part that fails to conform to the warranty during the warranty period.

Terms and Conditions

The warranty applies for a period of two years (for unlimited hours of usage on the unit during the two years), beginning on the date that the Dealer from which the unit is purchased submits the warranty registration form to LeeBoy (typically on the date of purchase).

- The warranty is not transferable and applies only to the original purchaser of a new unit from a Dealer.
- Warranty work must be performed at an authorized Dealer location.
- LeeBoy has the right to repair any component or part before replacing it with a new one.
- Replacement parts furnished by LeeBoy are covered for the remainder of the warranty period applicable to the unit or component in which such parts are installed.
- All new replacement parts purchased from a Dealer will carry a six (6) month warranty.

Exclusions and Limitations

LeeBoy has no obligation under this warranty for any defects or failures caused by:

- Misuse, misapplication, negligence, accident or failure to maintain or use in accordance with the most current operating instructions;
- Unauthorized alterations or modifications;
- Any replacement parts or attachments not manufactured by or expressly approved by LeeBoy;
- Failure to conduct normal maintenance and operating service, including without limitation, providing lubricants, coolant, fuel, tune-ups, inspections or adjustments; or
- Unreasonable delay, as determined by LeeBoy, in making the applicable units or parts available upon notification of a service notice ordered by LeeBoy.

LeeBoy is not responsible for any of the following:

- Defects or failures with respect to any engine or truck chassis;
- Defects or failures covered by a warranty provided by a third-party, including without limitation warranties with respect to engines or truck chassis;
- Costs related to travel time, mileage or overtime;
- Costs related to transporting the unit to and from the place at which warranty work is performed;
- Airfreight charges related to transporting repair parts to the place at which warranty work is performed;
- Used units or used parts of any kind;
- Repairs due to normal wear and tear, or brought about by abuse or lack of maintenance of the equipment, except for premature failures;
- Attachments not manufactured or installed by LeeBoy; or
- Miscellaneous charges.

Exclusion of Other Warranties/Limitation of Liability

OTHER THAN THE EXPRESS WARRANTY PROVIDED IN SECTION A. ABOVE, LEEBOY GIVES NO EXPRESS, STATUTORY, OR IMPLIED WARRANTY OR GUARANTEE AND HEREBY DISCLAIMS ALL SUCH WARRANTIES, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, SUITABILITY, DURABILITY, CONDITION, QUALITY, OR FREEDOM FROM CLAIMS OF ANY PERSON BY WAY OF INTERFERENCE, INFRINGEMENT, OR THE LIKE.

IN NO EVENT OR UNDER ANY THEORY OF RECOVERY, WHETHER AS A RESULT OF BREACH OF CONTRACT OR WARRANTY, ALLEGED NEGLIGENCE, OR LIABILITY WITHOUT FAULT, SHALL LEEBOY BE LIABLE FOR ANY SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE SALE, USE, OR LOSS OF USE OF ANY PRODUCT, INCLUDING WITHOUT LIMITATION LOST PROFITS OR REVENUES, COST OF CAPITAL, COST OF SUBSTITUTED EQUIPMENT, FACILITIES OR SERVICES, DOWNTIME COSTS, LABOR COSTS, OR DIMINUTION OF VALUE.

CONTACT INFORMATION

For information regarding parts and repairs about your LeeBoy product, contact your authorized Rosco/ LeeBoy dealer. If your dealer is unable to resolve the problem, contact LeeBoy directly.

Record dealer information in the space provided. For additional information about LeeBoy, please visit: www.leeboy.com.

Sales Representative: _____

Dealership Name: _____

Dealership Address: _____

Dealership Phone: _____

Record of Ownership

Please complete the following information for use if you need to contact Rosco/LeeBoy for service, parts or literature.

Machine Model Number: _____

Product Serial Number: _____

Date of Purchase: _____

Nameplate

The nameplate contains the model and serial numbers used to identify the machine and its components for parts or service information. Refer to the Engine Operator's Manual for the location of the engine nameplate.



Figure 2-1. Nameplate Location

SPECIFICATION CHARTS

The specifications provided in this section include the machine weights, dimensions, performance, and torque values for both metric and standard inch fasteners.

⚠CAUTION **Replace original equipment only with components approved by Rosco/LeeBoy.**

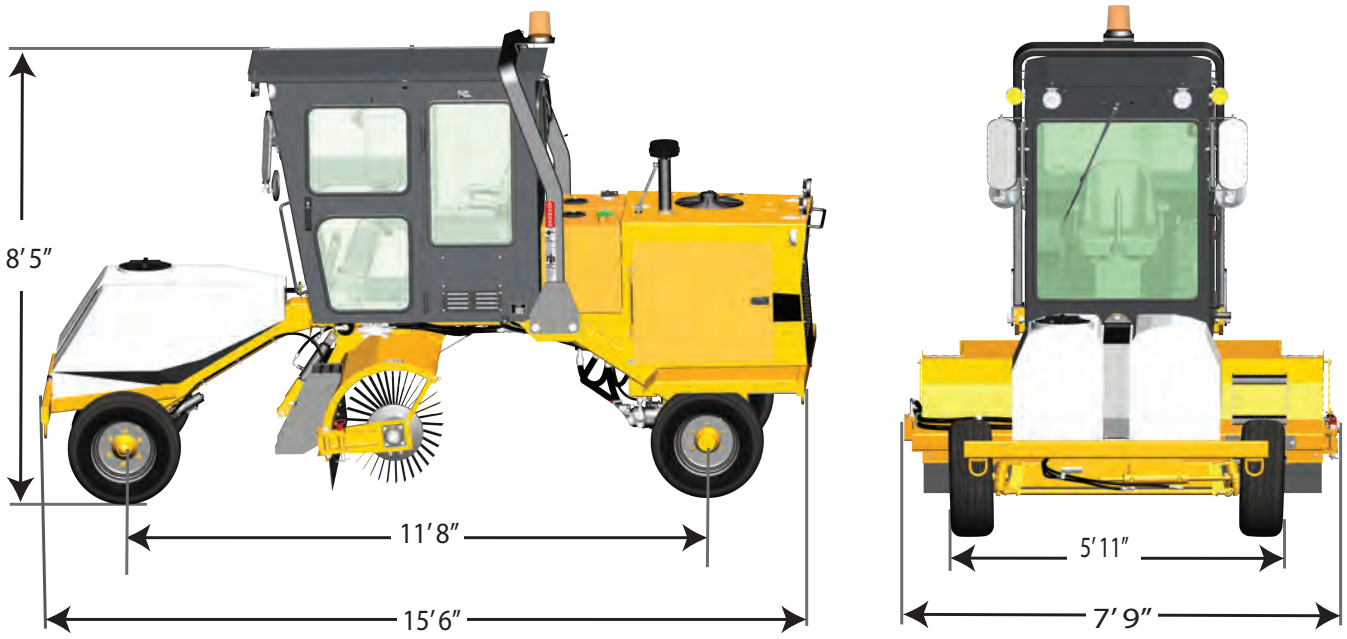


Table 2-1. Machine Dimensions

ITEM	SPECIFICATION
Overall Length	15 ft 6 in
Overall Height (with Roll-Over Protection System [ROPS])	8 ft 5 in
Overall Height (with ROPS and Beacon Light)	9 ft 5 in
Overall Width	7 ft 9 in
Tread Width (Front and Rear)	5 ft 11 in
Overall Weight (without Water)	5250 lb
Wheelbase	11 ft 8 in
Outside Turning Radius	18 ft 9 in
Ground Clearance	8.75 in
Cab	Includes: 12-Volt, Variable Speed Rear Defroster Fan; 12-Volt, 2-Speed Windshield Wiper; Windshield Washer (Front and Rear). Climate-Control Cab Options: 16,000 BTU/Hr (4690 Watt) Air Conditioning; 28,000 BTU/Hr (8206 Watt) Heater

Table 2-2. System Capacity Specifications

ITEM	SPECIFICATION
Fuel	30 gal
Engine Lubrication Oil Refill Capacity	7.4 qt
Engine Coolant	14 qt
Hydraulic Oil Tank	25 gal
Axle	4.7 qt

Table 2-3. Fuel and Lubricant Specifications

ITEM	SPECIFICATION
Engine Oil	15W-40, API CH-4, CI-4
Hydraulic Oil	All Weather All Temperature VG32
Fuel	Ultra-Low Sulfer Diesel
Grease	Shell Avania EP Grease or Equivalent
Drive Axle	80W-90LS GL5
Engine Coolant	Ethylene/Glycol or Propylene Glycol
Master Cylinder	All Weather All Temperature VG32

Table 2-4. Engine Specifications

ITEM	SPECIFICATION
Manufacturer and Model	Hatz 4H50TIC
Engine Type	74 HP, 4-Cycle Liquid Cooled Diesel, Tier 4 Final
Peak Torque	184 ft. lb. @ 1600 RPMs
Displacement	119 cu in
Combustion System; Intake System	Direct Injection; Turbocharged, Air Cooled
Power Rating HP (kW)	74 HP @ 2500 RPMs
Maximum Speed	2500 RPMs
Fuel	Low Sulfer Diesel (do not use any other type of fuel)

Table 2-5. Drive System Specifications

ITEM	SPECIFICATION
Transmission	2-Speed Hydrostatic
Steering	Hydraulic, Orbital Motor, 3.7 GPM. Priority flow at 103 Bar, 1500 PSI.
Rear Axle	Comer 068 Series
Tire Size	ST225x75-15R, Load Range E
Tire Inflation Pressure	50 PSI, 3.45 Bar (345 kPa)
Travel Speed	LOW: 0 - 12 mpg HIGH: 0 to 22 mpg
Hydrostatic Pump Displacement	2.8 CIR
Hydrostatic Motor Displacement	2.8 CIR
Brakes, Limited Slip	2-Wheel Wet Disc, Foot-Operated with Hand-Operated Park Brake

Table 2-6. Electrical Specifications

ITEM	SPECIFICATION
Battery	12 Vdc Maintenance Free, Negative Ground
Cold Cranking Amps (CCA)	770 CCA
Alternator Output Amperage	110 Amps @ 2500 RPMs
Alternator Fan Belt Tension	Manual belt tension mechanism keeps serpentine belt under tension at all times.

Table 2-7. Brush Operating System

ITEM	SPECIFICATION
Brush Angle	45° Left or Right
Brush Speed	200 RPMs maximum (variable with engine RPMs)
Brush Core (Welded Steel)	7 ft 6 in x 32 in, OD Poly Bristles
Brush Down Pressure Control	Hydraulic, Fully Adjustable
Brush Cover (Steel)	Full Length: 160°
Brush Watering System	Poly Tank, 1 @ 150 gal. Strainer, 12-Volt Diaphragm Pump
Motor	Direct Drive
Pump (Gear)	19 GPM, 207 Bar, 3000 PSI

Table 2-8. Miscellaneous Standard Equipment

ITEM	SPECIFICATION
Horn	12-Volt, Automotive
Back-Up Alarm	97 db +/- 4 db at 4 ft
Lights	Turns Signals, Tail Lights, Work Lights, Stop Lights, Beacon Light
12-Volt Outlet	12-Volt Electrical Outlet Connection
Suspension Seat	Suspension Adjustable with Seat Belt

Table 2-9. Optional Equipment

ITEM	SPECIFICATION
Dust Suppression System	150-Gal (568 L) Water Tank; Spray Bar with Corrosion-Resistant Spray Nozzles
Brush Filler (Water or Tube Type)	7 ft 6 in x 32 in
Strike-Off Blade	Front-Mounted
Curb Brush	Side-Mounted (Front Right Bumper)
Radio	CD/Weather Band Radio
Back-Up Camera	LCD Color Monitor

TORQUE SPECIFICATIONS

The following tables list torque values for standard hardware. This is a guide for average application involving typical stresses and machined surfaces. Values are based upon physical limitations of clean, plated and lubricated hardware. Under more extreme conditions, individual torque value should be followed.

Conversion formulas are provided below:

Conversion	Formula
ft-lb to N•m	[ft-lb]*1.3558 = [N•m)
ft-lb to in-lb	[ft-lb]*12 = [in-lb]
N•m to in-lb	[N•m]*8.8508 = [in-lb]

Standard Inch Fasteners

Table 2-1.Torque Specifications For Standard Inch Fasteners

SIZE	THREAD	CAPSCREWS: SAE GRADE 5				CAPSCREWS: SAE GRADE 8			
		TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
		Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
1/4	20 UNC	8	6	11	8	12	9	16	12
	28 UNF	10	7	14	9	14	10	19	14
5/16	18 UNC	17	13	23	18	25	18	34	24
	24 UNF	19	15	26	20	27	20	37	27
3/8	16 UNC	31	23	42	31	44	33	60	45
	24 UNF	35	26	47	35	49	37	66	50
7/16	14 UNC	49	37	66	50	70	52	95	71
	20 UNF	55	41	75	56	78	58	106	79
1/2	13 UNC	75	57	102	77	106	80	144	108
	20 UNF	85	64	115	87	120	90	163	122
9/16	12 UNC	109	82	148	111	154	115	209	156
	18 UNF	121	91	164	123	171	128	232	174
5/8	11 UNC	150	113	203	153	212	159	287	216
	18 UNF	170	127	230	172	240	180	325	244
3/4	10 UNC	267	200	362	271	376	282	510	382
	16 UNF	297	223	403	302	420	315	569	427
7/8	9 UNC	429	322	582	437	606	455	822	617
	14 UNF	474	355	643	481	669	502	907	681
1	8 UNC	644	483	873	655	909	681	1232	923
	14 UNF	722	542	979	735	1020	765	1383	1037
1-1/4	7 UNC	1121	840	1520	1139	1817	1363	2464	1848
	12 UNF	1241	930	1683	1261	2012	1509	2728	2046
1-1/2	6 UNC	1950	1462	2644	1982	3162	2371	4287	3215
	12 UNF	2194	1645	2975	2230	3557	2668	4823	3617

Metric Fasteners

Table 2-12. Torque Specifications for Metric Fasteners

NOMINAL SIZE AND PITCH	CLASS 8.8 [GRADE 5 EQUIVALENT]				CLASS 10.9 [GRADE 8 EQUIVALENT]			
	TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
	Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
M4 x 0.7	2	2	3	2	3	2	4	3
M5 x 0.8	5	3	7	4	7	5	9	7
M6 x 1	8	6	11	8	11	8	15	11
M8 x 1.25	19	14	26	19	27	20	37	27
M10 x 1.5	37	28	50	38	53	40	72	54
M12 x 1.75	65	49	88	66	93	70	126	95
M14 x 2	104	78	141	106	148	111	201	150
M16 x 2	161	121	218	164	230	173	312	235
M18 x 2.5	222	167	301	226	318	239	431	324
M20 x 2.5	314	236	426	320	449	337	609	457
M22 x 2.5	428	321	580	435	613	460	831	624
M24 x 3	543	407	736	552	777	582	1053	789
M27 x 3	796	597	1079	809	1139	854	1544	1158
M30 x 3.5	1079	809	1463	1097	1544	1158	2093	1570

Hydraulic Fittings

Tightening Flare-Type Tube Fittings

1. Check the flare and flare seat for defects that might cause leakage.
2. Align tube with fitting before tightening.
3. Lubricate connection.
4. Hand tighten swivel nut until snug.
5. To prevent twisting the tube(s), use two wrenches. Place one wrench on the connector body and tighten the swivel nut with the second to the torque shown in the following table:

NOTE: The torque values shown are based upon lubricated connections.

Table 2-13. Torque Specifications for Steel Flare Type Tube Fittings

TUBE SIZE OUTER DIAMETER	NUT SIZE ACROSS FLATS	TORQUE VALUE	
(IN)	(IN)	(LB FT)	(N•m)
3/16	7/16	8	11
1/4	9/16	12	16
5/16	5/8	16	22
3/8	11/16	23	31
1/2	7/8	38	52
5/8	1	54	73
3/4	1 1/4	75	102
7/8	1 3/8	83	113

Determining Proper Torque

The only reliable method of creating a consistently leak-free and long-lasting connection is to ensure the coupling is brought to the proper torque. Using a torque wrench with crowfoot is the best method, but the flats method can be used if a torque wrench is not available.

The most straightforward method of determining the correct torque setting is to multiply the desired torque by the length of the wrench from the center of the handle to the center of the drive (L); divided by the length of the wrench from the center of the handle to the crowfoot center (LA) as shown below:

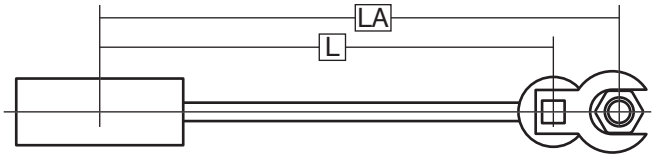


Figure 2-2. Torque Wrench - Crowfoot

NOTICE **The minimum torque values are adequate for sealing most applications. Maximum torque values should never be exceeded.**

There are several methods of determining the correct setting on the torque wrench when using a crowfoot. All of the methods involve making the setting proportional to the effective change in length of the wrench multiplied by the desired final torque. The equations and illustration below describe proper measurements.

Equations

- Torque setting if the crowfoot is placed in line with respect to the wrench:

TS = TD * L / LA

OR

TS = TD * L / (L+E)

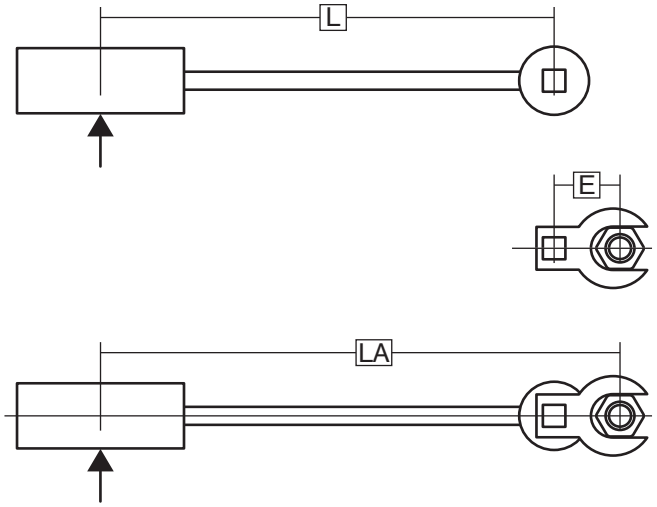


Figure 2-3. Measurements Needed

LEGEND

L = Distance from center of torque wrench handle to the center of socket drive

E = Distance from center of socket drive to the center of crowfoot

LA = Distance from center of torque wrench handle to the center of crowfoot

TD = Desired torque at the fitting

TS = Torque setting indicated on wrench

Coupling Installation

Use the following steps for proper coupling installation:

1. Determine the correct torque value for your coupling.

NOTE: Only use the torque values specified from the manufacturer. DO NOT use SAE torque recommendations.

2. Ensure the seal face and threads are clean and in good condition. O-Rings should be lubricated with light oil, but threads should be completely dry unless making pipe thread connections (interference seal).

NOTE: Attach the male end of the hose onto the equipment first since it may be necessary to rotate the entire hose assembly to tighten the male threads. Then route the hose into position while avoiding twisting the hose.

3. Tighten the connection (by hand), bringing the seal face into contact and rotating the nut until it stops.
4. Mark a line across the coupling nut and backup hex for the flats method verification of coupling torque.
5. Apply a wrench to the backup hex to prevent the coupling and hose from moving while tightening the nut with a torque wrench.

NOTICE **Failure to retain the backup hex during installation will also result in additional clamp load force that could cause damage to the seal face.**

NOTE: The coupling nut must be in motion for an accurate torque reading. If the nut is stopped before final torque value is achieved, it must be loosened and retightened until the torque is attained while the nut is in motion.

If a torque wrench cannot fit into the coupling area or if it is unavailable, the flats method may be used to ensure that the coupling is properly tightened.

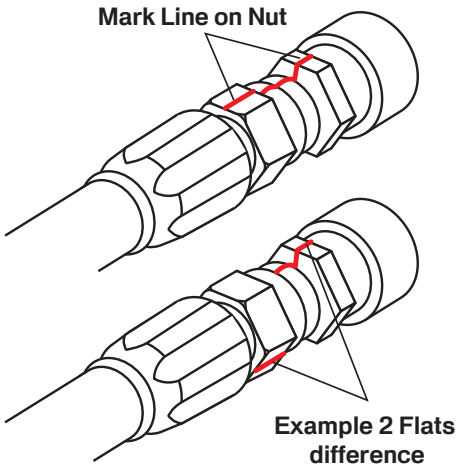


Figure 2-4. Flats Method Tightening

NOTE: The mark placed on the nut and backup hex after tightening by hand should rotate during final tightening according to the table below. The nut and backup hex can then be marked to indicate if the coupling loosens over time.

Table 2-14. Flats Method Values for Selected Terminations

FLATS METHOD VALUES		
Termination Type	Dash Size	Flats
JIC	-4	1.5 - 1.75
JIC	-6	1.0 - 1.5
JIC	-8	1.5 - 1.75
JIC	-10	1.0 - 1.5
JIC	-12	1.0 - 1.5
JIC	-16	.75 - 1.0
JIC	-20	.75 - 1.0
JIC	-24	.75 - 1.0
JIC	-32	.75 - 1.0
JIS	-4	.5 - 1.5

1. Seal faces must be in contact with the fitting fully tightened by hand before marking flats.
2. The flats method is most accurate for the first assembly cycle. For multiple disassembly and assembly cycles, torque values are more reliable.
3. Tightening two (2) flats or more may damage seal faces.

Table 2-15. Torque Specifications For US Style Coupling Terminations

JIC, SAE 45°, ORFS, O-RING BOSS, GATES ADAPTERLESS AND MEGASEAL										
DASH SIZE	JIC 37°, SAE 45° & Mega-Seal (Steel)		JIC 37°, SAE 45° & Mega-Seal (Brass)		Flat Face O-Ring Seal (Steel)		SAE O-Ring Boss (Steel) & Gates Adapterless ≤ 4000 PSI		SAE O-Ring Boss (Steel) & Gates Adapterless > 4000 PSI	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
-3									8	10
-4	10	11	5	6	10	12	14	16	14	16
-5	13	15	7	9					18	20
-6	17	19	12	15	18	20	24	26	24	26
-8	34	38	20	24	32	40	37	44	50	60
-10	50	56	34	40	46	56	50	60	72	80
-12	70	78	53	60	65	80	75	83	125	135
-14					65	80			160	180
-16	94	104	74	82	92	105	111	125	200	220
-20	124	138	75	83	125	140	133	152	210	280
-24	156	173	79	87	150	180	156	184	270	360
-32	219	243	158	175						

Table 2-16. Torque Specifications for DIN 24, DIN 60, and Inverted Cone Style Coupling Terminations

DIN 24, DIN 60, AND INVERTED CONE			
Size (mm)		Torque (lb ft)	
Light Series Tube OD	Heavy Series Tube OD	Min	Max
6		7	15
8		15	26
10	8	18	30
12	10	22	33
14	12	26	37
15	14	30	52
	16	30	52
18	20	44	74
22	25	59	89
28	30	74	111
	38	74	162
35		133	184
42		148	221

Table 2-17. Torque Specifications for 4-Bolt Flange Connections

Figure 2-1. 4-BOLT FLANGES		
Dash Size	Bolt Size (in)	Torque (lb ft)
-8	0.31	17
-12	0.38	26
-16	0.44	43
-20	0.50	65
-24	0.63	130
-32	0.75	220

1. Align faces and tighten bolts (by hand) before applying final torque in a pattern. The seal faces must be parallel with an even bolt tension to seal properly.
2. Torque values apply to bolts that are plated or coated in light engine oil.
3. Before assembly, lubricate O-Ring with light oil (SAE 10W or 20W).

Table 2-18. Torque Specifications for NPTF Dry Seal Pipe Threads

Figure 2-1. NPTF	
Dash Size	**Max Torque (ft-lb)**
-2	20
-4	25
-6	35
-8	45
-12	55
-16	65
-20	80
-24	95
-32	120

1. The torque values obtained from tightening pipe threads can vary considerably depending upon thread condition. Adequate sealing can occur at values much lower than the maximum values listed above. Only enough torque to achieve adequate sealing should be used.
2. When using a male tapered pipe thread with a female straight or parallel pipe thread, maximum values are 50% of those listed in the table above.
3. If thread sealant is used, maximum values shown should be decreased by 25%.

Table 2-19. Torque Specifications for BSP 30° Inverted Cone and JIS Coupling Terminations

BSP 30° INVERTED CONE AND JIS		
Dash Size	**Torque (ft-lb)**	
	Min	**Max**
-2	7	9
-4	11	18
-6	19	28
-8	30	36
-10	37	44
-12	50	60
-16	79	95
-20	127	152
-24	167	190
-32	262	314

Section 3

COMPONENTS

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NOTES

COMPONENTS OVERVIEW

This section describes the major components for the Rosco RB50 Broom. Become familiar with these components before operating the machine.



Broom Cab

The Rosco RB50 Broom is equipped with a fully-enclosed, pressurized cab for maximum operator comfort on the job site, including air conditioning and heat. Cab options are also available.

The broom is equipped with the PV380 Digital Display on the control panel to the right of the operator that features throttle control, tachometer, fuel gauge, oil pressure gauge, coolant temperature, voltmeter and hour meter.

The main control panel is conveniently located to the right of the operator's seat. Various controls and a joystick equipped with controls for floating brush and swinging the broom left or right enhance efficiency and ease of operation on the job site.

Engine

The Rosco RB50 Broom uses a four-cylinder, turbo-charged diesel engine to drive the hydraulic auxiliary pumps for steering and broom control. The Hatz 74 HP turbo diesel engine performs at 2500 RPMs, and is easily accessible from either side of the machine.



An air cleaner mounted on top of the engine filters fine particles such as dust, sand, chaff, and lint from the air. A pre-cleaner removes larger particles of dirt and debris before the air enters the air filter elements, depositing the trapped materials into the pre-cleaner bowl.

As air is taken in, a cyclone action deposits some of the fine particles in the evacuator mounted on the bottom of the air cleaner housing. The evacuator is held closed during engine operation by suction. This pre-cleaner system relieves the load on the air filter elements and allows longer intervals between servicing.

Primary and secondary fuel filters remove contaminants from diesel fuel before the fuel flows to the injection pump for injection into the engine combustion chamber.

A radiator mounted at the rear of the broom cools the engine. As coolant flows through the radiator, airflow from the engine-driven fan removes heat from the coolant.

Refer to your Engine Operator's Manual for a complete description and specifications of the engine.

Brush Control and Quick-Change Core System

The brush drive speed is regulated by the engine speed and a control dial on the main control panel. Brush lift and brush swing are controlled by push buttons and control dial on the control panel. The buttons operate solenoid-controlled hydraulic valves.



The Quick-Change Core system features a single-motor drive that is suited for applications ranging from road construction and industrial site cleanup to landscaping and snow removal. The quick-change core requires no tools to change the brush.

Electrical System

The electrical system is powered by a 12-volt battery in the engine compartment. The battery maintains 770 cold cranking amperes (CCA). The battery charge rate can be monitored using the voltmeter included in the PV380 Digital Display unit.

An engine-mounted alternator capable of 110 amperes charging capacity keeps the battery charged during normal operation.

Hydrostatic Drive System

The Rosco RB50 Broom is a hydrostatic, self-propelled unit. The hydrostatic drive and hydraulic systems power the steering and broom drives using one common hydraulic oil tank.

This system propels the RB50 Broom forward and reverse, which also acts as dynamic braking. A variable displacement pump provides power to a fixed displacement two-speed motor that drives the rear axle.

The operator controls the drive system via a joystick on the control panel located on the right side of the seat. A foot pedal transmission option is also available. A neutral position, which activates a neutral start switch, is included between forward and reverse directions of the control lever. A toggle switch engages high or low speed.

A hydraulic pump connected to the hydrostatic system pump provides hydraulic flow to the manifold. This manifold provides priority flow to the steering system and the brush lift-and-swing valves at approximately five (5) gallons per minute. Excess flow from the pump provides power for the brush drive motor. The steering flow always receives the priority flow, therefore, it is always powered.

Steering is performed by a hydraulically-powered orbital system operated by a steering wheel.

MACHINE OVERVIEW

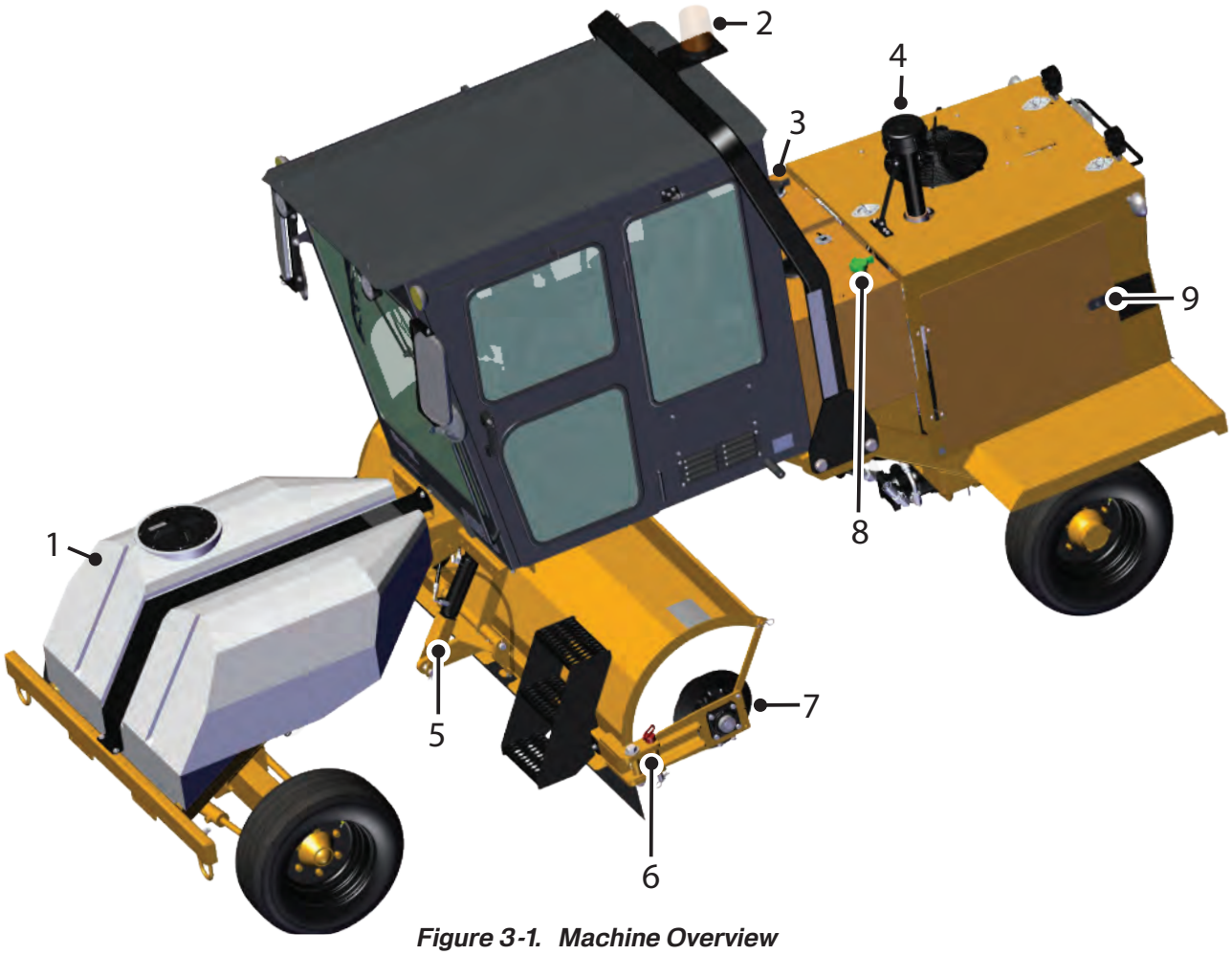


Figure 3-1. Machine Overview

ITEM NO.	COMPONENT NAME	FUNCTION
1	Water Tank (Option)	Water tank mounted on front of machine if equipped with optional Dust Suppression System. Holds 150 gallons.
2	Beacon Light	Flashing beacon light for added safety to alert traffic of broom operation. The beacon light is removable.
3	Hydraulic Oil Filler Cap	Hydraulic oil filler cap contains strainer to prevent debris from entering system when filling hydraulic tank.
4	Air Intake Stack	Rotary pre-cleaner.
5	Brush Lock Arm	Manual lock arm for locking brush into place (position shown) during travel and transport.
6	Brush Core Lock/Release	Lock pin releases and locks brush core into place when changing brush. No tools are required to change brush.
7	Brush	Angles 45 degrees right or left. 32-inch steel brush tube with many different bristle options.
8	Fuel Filler Cap	Fill using only ULSD (Ultra-Low Sulfur Diesel) fuel.
9	Engine Access Panel	Locking panel to access engine (on each side of the machine).

RB50 BROOM CONTROLS

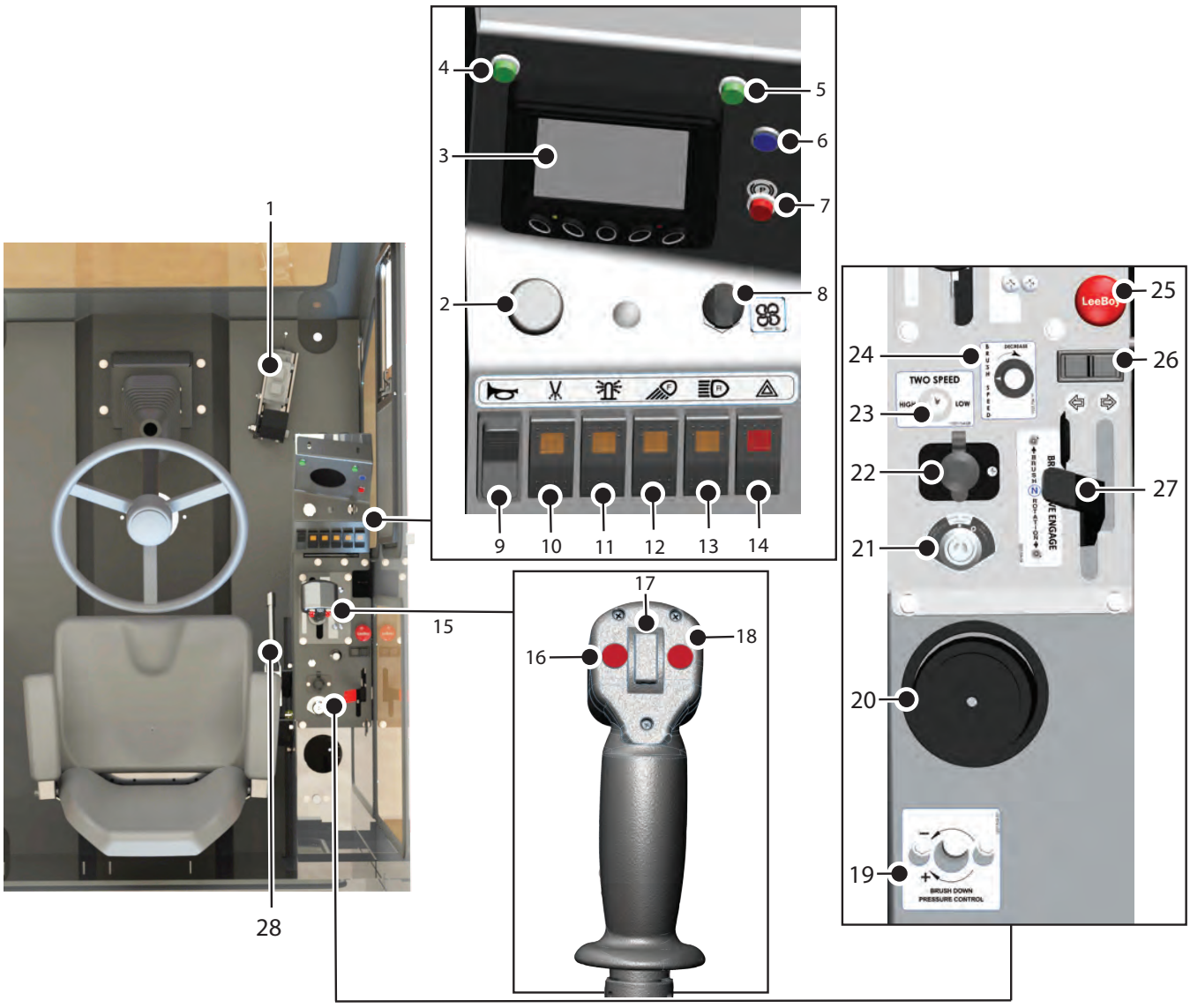


Figure 3-2. RB50 Controls

ITEM NO.	COMPONENT NAME	FUNCTION
1	Foot Brake	GENTLY step on brake to stop the broom.
2	Windshield Wiper/Washer	Turn knob to active wipers. Press knob to spray washing fluid.
3	PV380 Digital Display	Multi-functional display that shows engine temperature, voltage, oil pressure, engine RPMs, machine hours, service reminders, diagnostics, and are alerted to any fault conditions that require attention.
4	Left Signal Light	Illuminates when left turn signal is engaged.
5	Right Signal Light	Illuminates when right turn signal is engaged.
6	High Gear Light	Illuminates when High Gear is engaged.

ITEM NO.	COMPONENT NAME	FUNCTION
7	Park Brake Light	Illuminates when the park brake is engaged.
8	Cab Fan	Turns cab fan on and off. Turn button right to activate.
9	Horn	Press button to sound horn.
10	Water System	Turns the water system on and off.
11	Beacon Light	Turns beacon light on and off.
12	Front Work Lights	Turns front work lights on and off.
13	Rear Work Lights	Turns rear work lights on and off.
14	Hazard Lights	Flashing hazard lights to alert traffic.
15	Transmission Selector Lever / Joystick	Pushing joystick forward increases machine speed while pulling joystick backward slows machine speed. Also features controls for floating and swinging brush left or right.
16	Brush Swing Left	Controls brush swing to the left.
17	Brush Lift/Float	Controls brush lift and float.
18	Brush Swing Right	Controls brush swing to the right.
19	Brush Down Pressure	Controls the brush pressure on the work surface. Turn clockwise to increase pressure and counterclockwise to decrease pressure.
20	Cup Holder	The cup holder is an added convenience for the operator.
21	Ignition Switch	Turn key clockwise to start the engine. Turn key counterclockwise to shut off the engine. Transmission Selector MUST be in the NEUTRAL position to start. Features a Neutral Start system to prevent operation of the engine starter when the transmission is not in neutral.
22	12-Volt Outlet	12-volt electrical outlet.
23	Two Speed Switch	High Gear: Full displacement that allows higher transport speeds. Push switch left for High Gear. Low Gear: Partial displacement providing greater torque and lower speeds. Push switch right for Low Gear. **NOTE: Use ONLY low gear while sweeping.**
24	Brush Speed Dial	Turn dial left to increase brush speed. Turn dial right to decrease brush speed based on engine RPMs.
25	Emergency Stop	This button breaks power to the ignition switch. Push button to lock. Twist button to unlock.
26	Signal Lights	Push button left for left turn signal. Push button right for right turn signal.
27	Brush Drive Engage Lever	Turns brush rotation on and off. Push lever forward to engage. Pull lever backward to reverse brush rotation.
28	Park Brake Lever	Raise to engage park brake. Lower lever to release park brake.

FOOT PEDAL CONTROLS OPTION

The optional forward and reverse foot pedal in lieu of joystick transmission control allows for a broader range of operator driving opportunities. ***(See Page 4-22)*** Switches on the main control panel replace button controls on the joystick while the other control panel items remain the same as shown on ***Pages 3-6 and 3-7.***

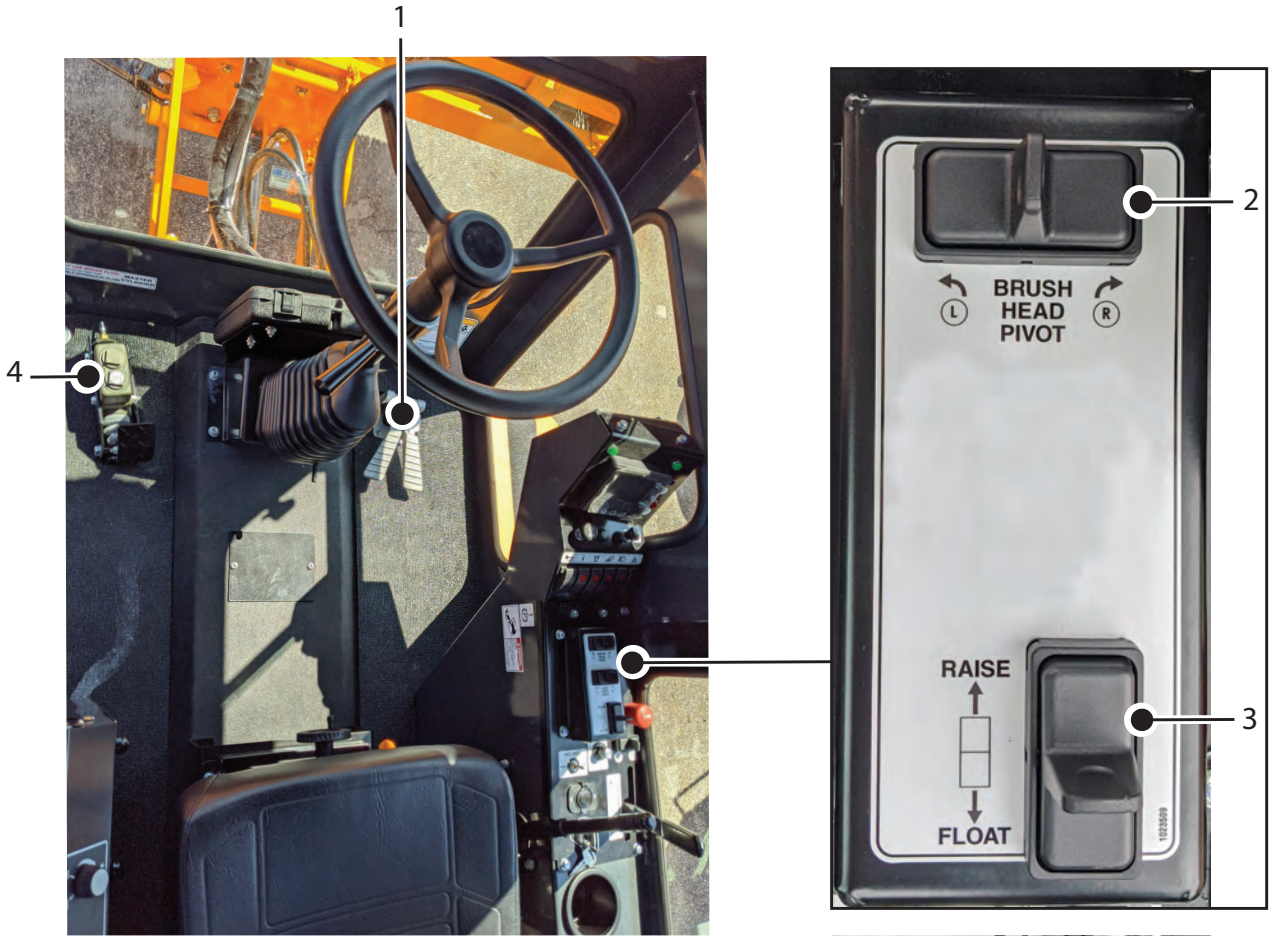


Table 3-3. Foot Pedal Controls Option

ITEM NO.	COMPONENT NAME	FUNCTION
1	Forward/Neutral/Reverse (FNR) Pedal	• Press the right pedal to move forward. • Press the left pedal to move in reverse. • The pedals automatically return to neutral when released.
2	Brush Head Pivot	Push the brush head pivot switch Left to swing the brush left; or right to swing the brush right.
3	Raise/Float	Push the Raise/Float button forward (raise) or back (float) for brush raise and float control.
4	Brake Pedal	Starting: Firmly press pedal to start machine. Braking: GENTLY press pedal to stop machine. The pedal is very sensitive.

Section 4

OPERATION

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GENERAL INFORMATION

The Rosco RB50 Broom is a hydrostatic, self-propelled unit. The broom angles at 45 degrees right or left for efficient general road sweeping or snow removal. The hydraulically-controlled RB50 Broom can brush, scour and remove job site debris with ease.

The RB50 features a Quick-Change Core system with a single-motor drive that is suited for applications ranging from road construction and industrial site cleanup to landscaping and snow removal. The quick-change core requires no tools to change the brush.

The Rosco RB50 is ideal for cleaning parking lots, industrial plants, airport runways, streets and driveways. Because the RB50 can clean large surfaces quickly and efficiently due to its bristle rotation, ability to angle and adjust the brush-down force, it is perfect for a variety of construction, industrial and landscaping applications.

Options include a fully-enclosed cab version, curb brush and snow blade attachments. To loosen particles and reduce dust, an optional Dust Suppression System is available, equipped with its own electric pump, a 150-gallon water tank, corrosion-resistant spray nozzles and piping system.

⚠ DANGER Only authorized personnel who are properly trained should operate this machine.

Receiving the Machine

Although the machine has been checked thoroughly by the manufacturer, road hazards or other factors during transport may result in damage. Inspect machine for any damage during transport. Contact your authorized dealer immediately if any damage has occurred.

The RB50 Broom is equipped with ROPS (Roll-Over Protective System). The unit is shipped with ROPS in the stowed position. **The ROPS will need to be placed in its operating position prior to use. See the following page for the procedure to rotate ROPS from shipping to its operational position.**

Be sure to check the machine thoroughly upon receipt:

1. Rotate ROPS from the shipping position to the operational position. ***(Page 4-4)***
2. Check engine oil. ***(Page 5-9)***
3. Check hydraulic fluid level. ***(Page 5-18)***
4. Check machine hoses for any damage or leaks.

Initial and Daily Inspection

The following inspections are essential and should be performed before starting the machine. Visually inspect the unit to check its general condition. Continue with a check of the systems and components shown in **Table 4-1** below:

Table 4-1. Initial and Daily Inspection

INSPECT	PROCEDURE
Engine Oil Level	Maintain oil level between ADD and FULL marks on dipstick.
Fuel Tank	Check for adequate fuel supply. Fill if needed. Always fill tank at end of the day to prevent condensation in the tank.
Hydraulic Tank	Check the tank for oil level and leaks.
Radiator	Check coolant level.
Battery	Ensure all cables are tight and clean. Check for corrosion on the battery terminals.
Air Cleaner	Check the air filter element and connections. Air cleaner has both a primary and secondary filter.
Drain Plugs	Make sure plugs are inserted and tight.
Engine Belt	Check for proper belt tension.
Grease Fittings	Ensure fittings are greased and in good working order.

ROPS

The RB50 Broom is equipped with ROPS (Roll Over Protection System) as an added safety feature to protect the operator and equipment in the event of overturning the vehicle. The ROPS meets strict requirements designated by various agencies.

Use the following procedure for rotating ROPS from the shipping position to the working position:

1. Remove the bolts holding each of the two shipping straps (four bolts on each strap).
2. Loosen the two front ROPS mounting bolts. **DO NOT REMOVE. (Figure 4-1)**
3. Rotate the ROPS forward into position.
4. Retighten the mounting bolts and torque. ***(See Torque Tables in Section 2).***



Figure 4-1. ROPS in Shipping Position

START-UP PROCEDURE

A Neutral Start switch has been installed to prevent operation of the engine starter when the transmission is not in neutral.

You can check that it is functioning properly by:

1. Engage the park brake. **(Figure 4-2)**
2. Place the transmission selector into the NEUTRAL position.
3. Turn the ignition key to the START position. The Wait to Start light will illuminate for a few seconds on the PV380 digital display controller.
4. When the Wait to Start light goes off, turn key to the ON position.

NOTICE **Using starting additives, such as ether, is not recommended as severe engine damage can occur.**

⚠ DANGER **DO NOT leave the operator station unattended while the machine is in gear or in motion.**

⚠ CAUTION **Verify there are no people, obstacles or other equipment in the machine's path before starting the engine.**

To start the Rosco RB50 Broom:

1. Ensure the park brake is engaged before starting the machine.
2. Place the transmission selector lever into the NEUTRAL position.

⚠ WARNING **Be sure the park brake is engaged and transmission selector is in the neutral position before starting the engine. The machine's sudden movement if started in gear can cause serious injury or even death if operator is thrown from the machine.**

3. Insert key into the ignition and turn clockwise to the START position. The "Wait to Start" light will illuminate for a few seconds on the PV380 digital display controller. ***(Page 4-9)***
4. When the Wait to Start banner disappears, turn key to the ON position to crank engine.

NOTICE **DO NOT hold the starter longer than 10 - 15 seconds. If the engine does not start, allow the starter to cool two or three minutes between attempts to prevent damage to the starter.**

5. Set the engine throttle to one-half speed (1800 RPMs) using the PV380 digital controller. ***(Page 4-9)***